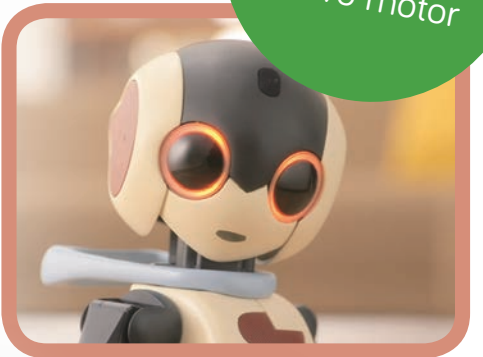
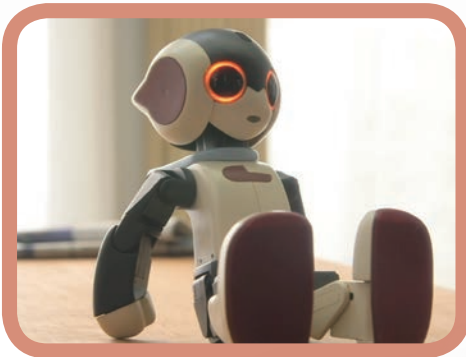
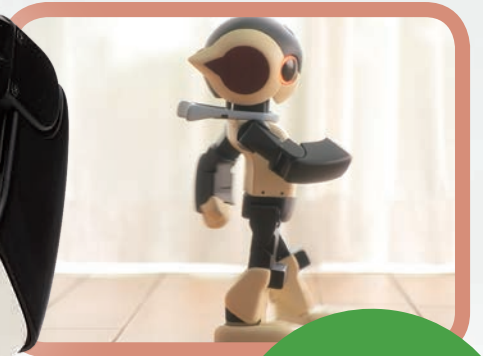


Build your own Robi

Pack 02



Stages 3-6: Fit and test Robi's first servo motor



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Assembly Guide

p14

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Stage 6: Testing Robi's head servo p27

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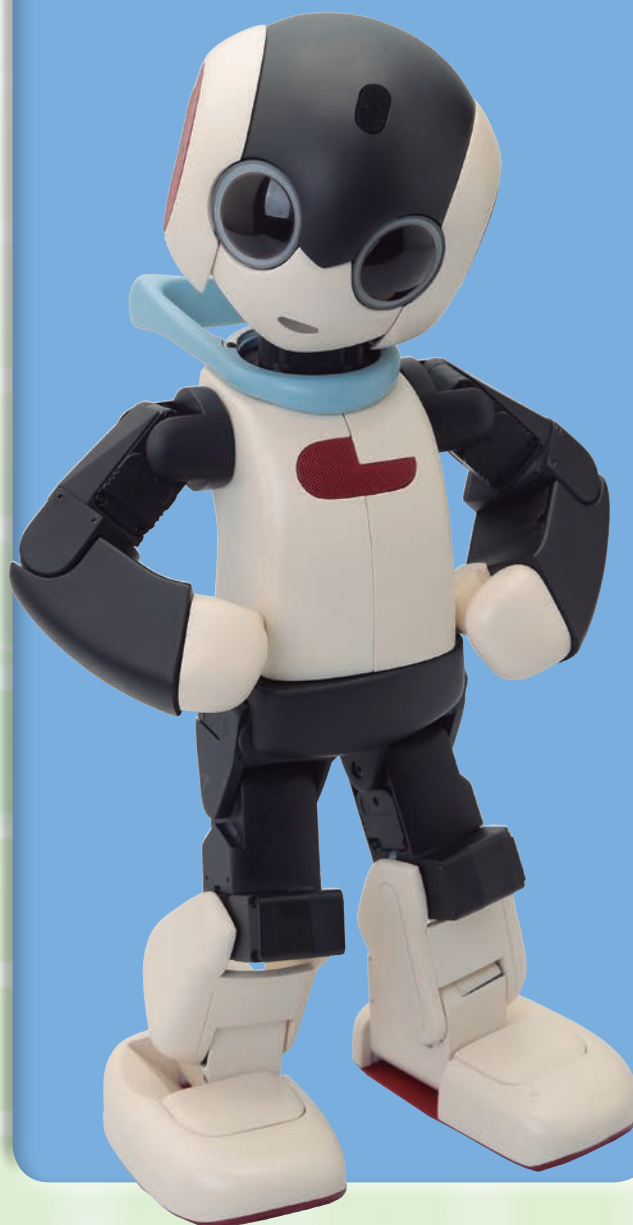
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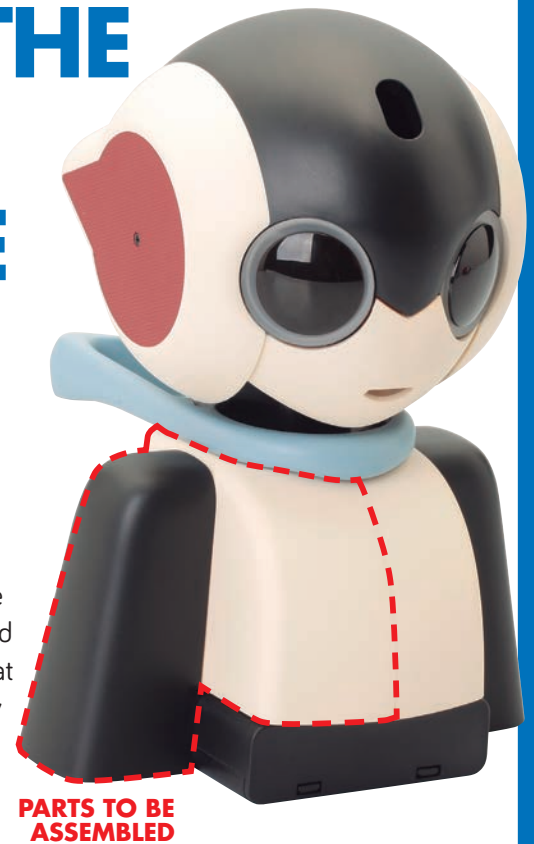


STAGE 3: PREPARE THE NECK STAND AND FIRST SERVO CABLE

In this session, you will begin assembling the neck stand that will be used to test Robi's head movement, as well as preparing the first of the servo cables.

By the start of Pack 2, you will have fitted the first servo that will move Robi's head, and will test this using the test board supplied with this stage. The neck stand is moulded to resemble Robi's upper body, and will support his head during the test to give you an accurate image of how the

parts will work on completion. You have also received your first servo cable, which you will fit with protective pads, ready for use. The test board and other parts supplied will not be used at this stage and should be stored safely until they are needed. Do not remove the test board from its plastic cover.



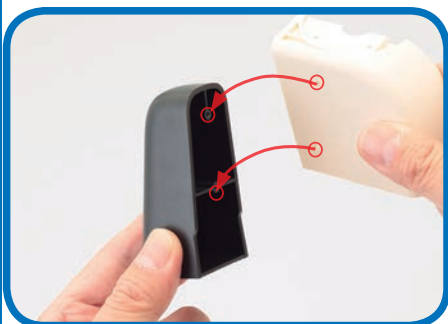
PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Scarf
- 2 Right shoulder (for neck stand)
- 3 Right chest (for neck stand)
- 4 Protective pads (50 pieces)
- 5 Test board
- 6 Servo cable (135mm)
- 7 M2 x 6mm pan-head screws x 3

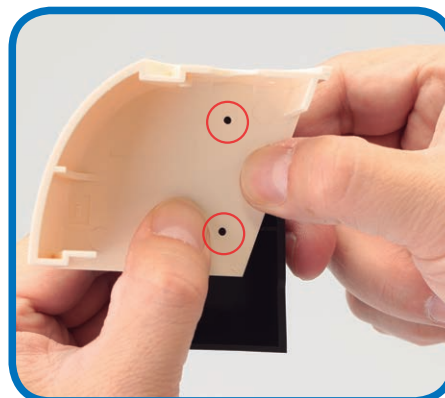
THE NECK STAND



- 1 Hold the right shoulder next to the right chest, as shown, to see where the parts will fit. The two vertically aligned holes on the smooth side of the chest will line up with the corresponding pair on the inside of the shoulder.



- 2 Hold the parts together so that the holes align.



- 3 Check from the inside of the chest to get a better view of the paired holes.

TIP!



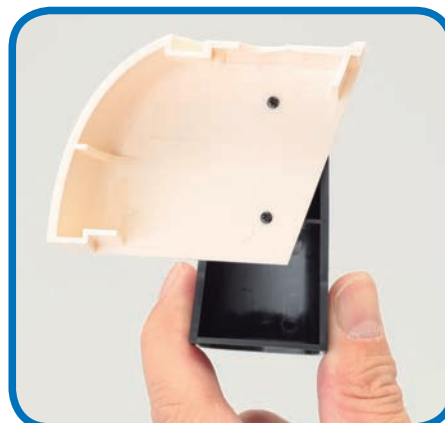
- 4 Place the first of the M2 x 6mm pan-head screws into the top hole from the inside of the chest, and tighten using the screwdriver supplied with Stage 2. Do not overtighten the screw.



If you find it difficult to locate the screw holes, try inserting the screw through from the inside of the chest first, and then joining the parts.

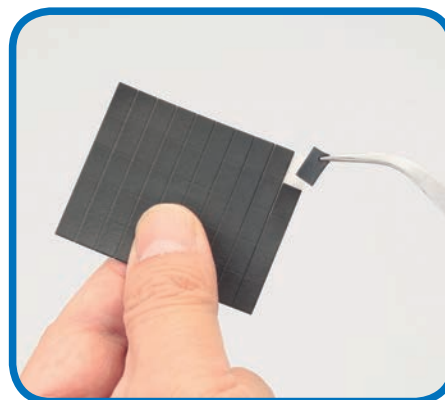
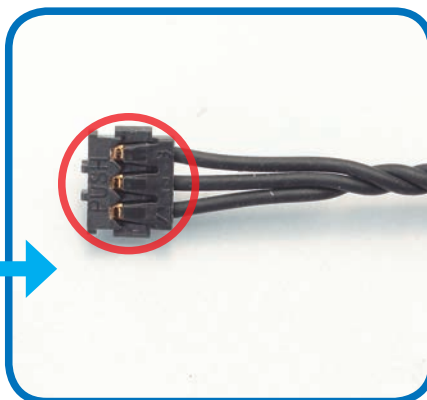


- 5 Place the second screw into the free hole, locate the corresponding hole on the inside of the shoulder, and tighten using the screwdriver.



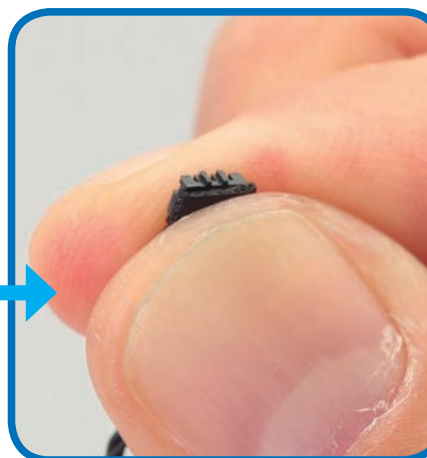
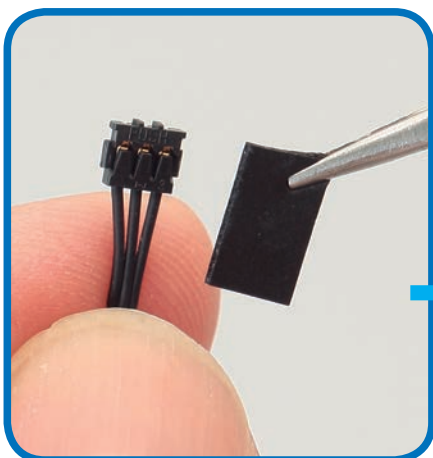
- 6 The first part of the neck stand should look like this.

SEALING THE SERVO CABLE



- 7** Inspect the connectors at either end of the servo cable. You will see that they are different on each side. You will apply the protective pad to the side with PUSH marked on it (circled).

- 8** Using tweezers, carefully peel a protective pad from the sheet.



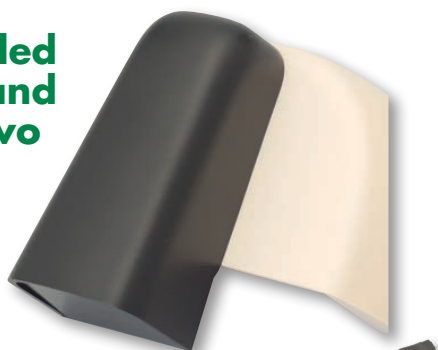
TIP!



The protective pad should be set centrally, and not extend beyond the tip of the connector.

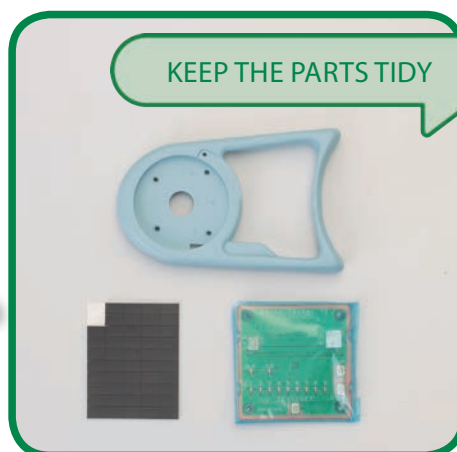
- 9** Holding the servo cable vertically with the side marked PUSH facing you, carefully pair it with the protective pad. Press the two together firmly between your finger and thumb to seal the connector. Once you have done this, repeat Steps 7-9 to seal the servo's other connector.

Assembled neck stand and servo cable



The first parts of the neck stand are complete, and your first servo cable is ready for use.

KEEP THE PARTS TIDY



Remember to store the unused parts safely until they are needed. Do not remove the test board from its sealed plastic cover, and be sure to keep all the protective pads, as they will be needed throughout the building process.

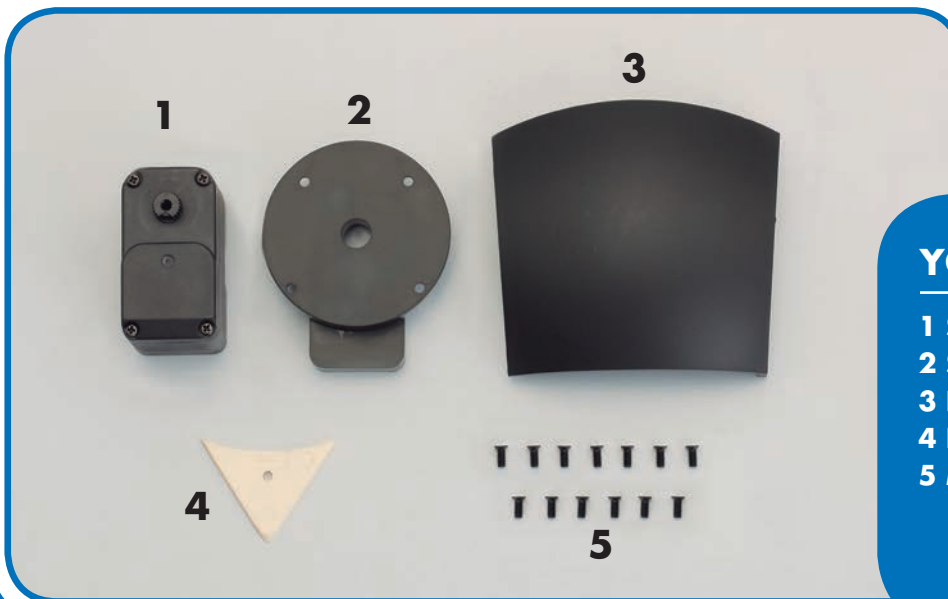
STAGE 4: ROBI'S HEAD, AND YOUR FIRST SERVO MOTOR

Continue building Robi's head, before modifying the first of his servo motors with a custom-built mounting plate.

In this stage, you will modify the servo motor used to move Robi's head. To do this, you will first remove the servo's factory-fitted cover and then replace it with a custom-made mounting plate specific to Robi. This will involve

opening the servo and briefly exposing the internal electronics board, so be especially careful at this stage of the assembly, as the electronics are very delicate and must be handled with extreme care.

PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Servo motor
- 2 Servo mounting plate
- 3 Back of head
- 4 Right ear backing
- 5 M2 x 4.5mm countersunk screws (x 13, two are spares)

SAVED PARTS

For this stage, you will need some of the parts you assembled or saved from previous stages. Be sure to have these to hand before proceeding.



Top of head (Stage 1)



Right ear base (Stage 1)



Head and face assembly (Stage 2)



Servo cable (Stage 3)

THE TOP OF ROBI'S HEAD



- 1** Lower the top of Robi's head from Stage 1 onto the existing head and face assembly, as shown. Make sure the pointed section at the front fits between his eyes.



- 2** Position the top of Robi's head so that the three holes circled in red line up with those on his left head frame.



- 3** Insert three of the M2 x 4.5mm countersunk screws into these holes and tighten them using the screwdriver supplied with Stage 2.



- 4** Turn Robi's head around, and repeat Steps 2 and 3 to attach the top of Robi's head to his right head frame.

THE BACK OF ROBI'S HEAD



- 5** Look closely at the moulding that forms the back of Robi's head. The left side features a short protruding flange with two closely spaced holes, and the right has a wider flange with two wider-spaced holes.



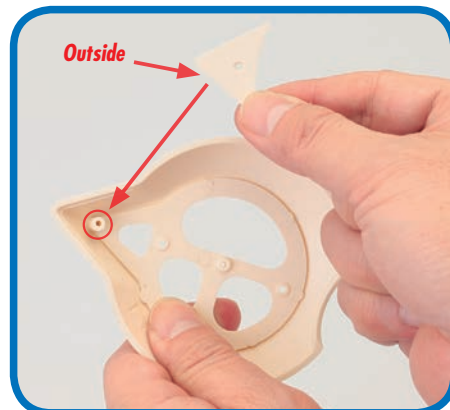
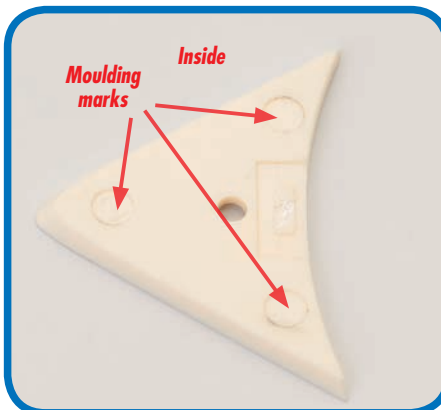
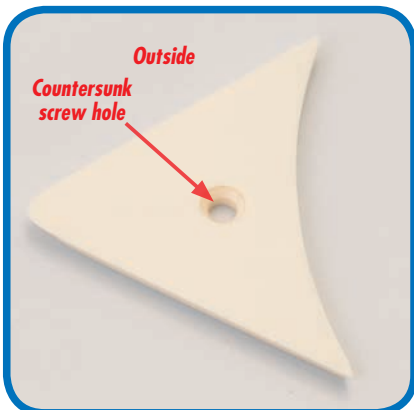
- 6** Fit the back of Robi's head against the top of his head so that the two closely spaced holes line up with the circled holes in the left head frame.



7 Fit two more M2 x 4.5mm countersunk screws into these holes and tighten them.

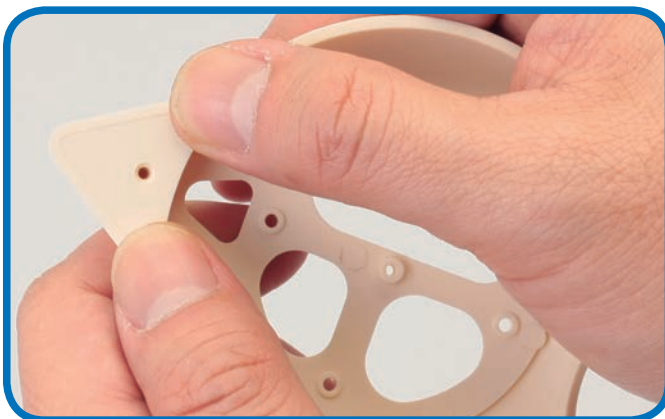
8 Turn Robi's head around and repeat Step 7 to attach the back of his head to the right head frame.

THE BACKS OF ROBI'S EARS

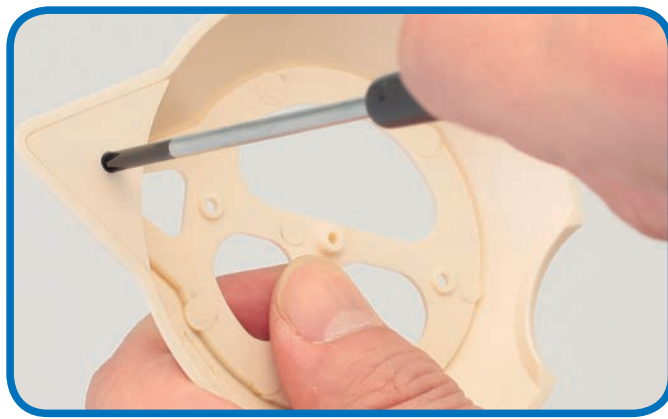


9 Inspect both sides of the right ear backing. The outside has a countersunk hole in its centre, into which a countersunk screw will fit, leaving the head flush with the surface of the part. The inside can be identified by the three circular marks left over from the plastic moulding process.

10 Take the right ear base supplied with Stage 1, and line up the ear backing, with the outside facing you.

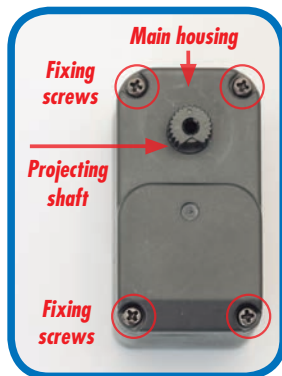


11 Press the ear backing into place, so that the countersunk hole lines up with the projection circled in Step 10.



12 Screw the parts together using another M2 x 4.5mm countersunk screw. The flat screw head should sit flush with the surface of the ear backing.

FITTING THE SERVO MOUNTING PLATE

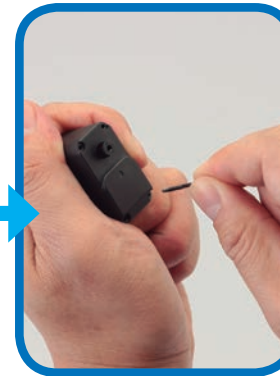


TIP!



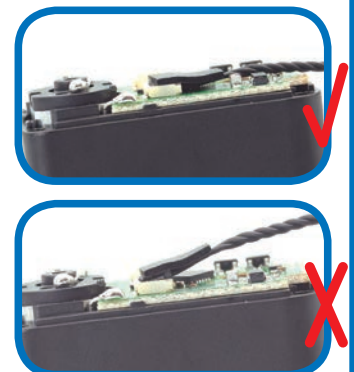
The electronics board under the servo's removable cover is extremely delicate, so be very careful not to touch any of the components with your fingers. To do this, always hold the servo housing on the sides, as shown.

13 Examine the servo to identify the removable cover. This is on the opposite face to the projecting shaft, and is held by four fixing screws (circled) that go right through the casing. You will see two gaps in the cover, beneath which are two connector sockets. The plug on the end of the servo cable that you fitted with a protective pad in Stage 3 will connect to the right hand one of these.



14 Carefully loosen all four screws, making sure not to use excessive force and damage the screw head. When about 5mm is visible, you will be able to undo them with your fingers. Keep these screws safely as you will use them to fit the mounting plate in place of the cover later.

15 Gently lift the cover away, taking care not to force it or damage the circuit board underneath.

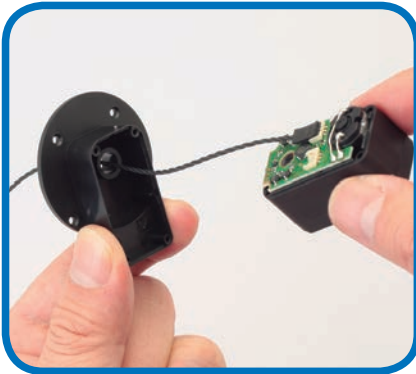


16 Place the servo on your work surface with the circuit board facing upwards. You will see two white connector sockets. The servo cable's connector will be fitted to the socket circled in red.

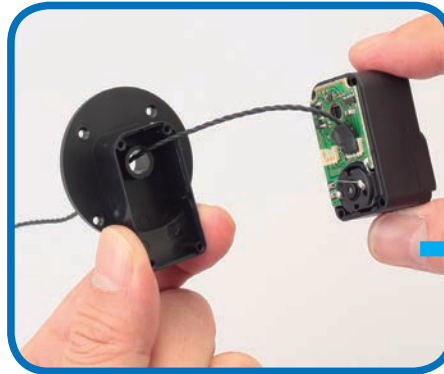
17 Carefully place the connector over this socket with the protective pad fitted in Stage 3 facing up. Once aligned, gently press on the pad until you feel the connector click into place.

18 Look closely at the connector from the top and the side to check whether it is attached correctly.

FITTING THE SERVO MOUNTING PLATE



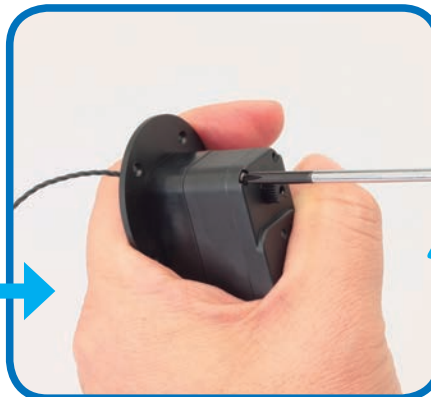
19 Hold the servo mounting plate as shown, and feed the free end of the servo cable through its central hole.



20 Fit the mounting plate onto the open face of the servo motor, feeding the surplus cable through the hole.



21 Take one of the four screws removed in Step 14 and tighten it back into its hole to secure the mounting plate to the servo.



TIP!



Lay the servo face down on your work surface if you find it difficult to tighten the screws when holding the parts in your hands.



22 Repeat for the remaining three screws to complete this stage of assembly.

Assembled head and servo



This stage is complete, so be sure to store the assembled parts, as well as any unused components, safely until next time.

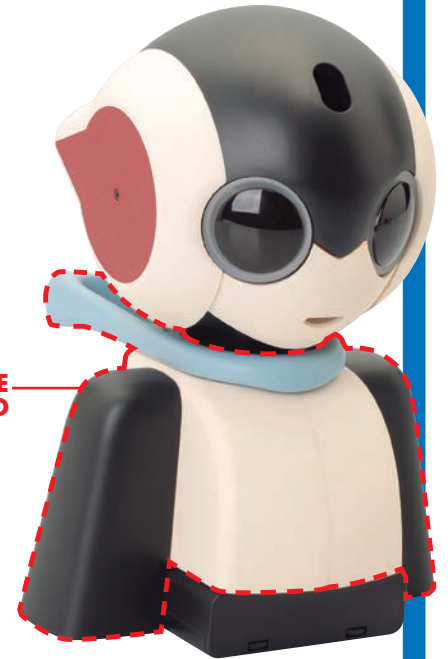


STAGE 5: ATTACHING ROBI'S SCARF AND HEAD SERVO TO THE NECK STAND

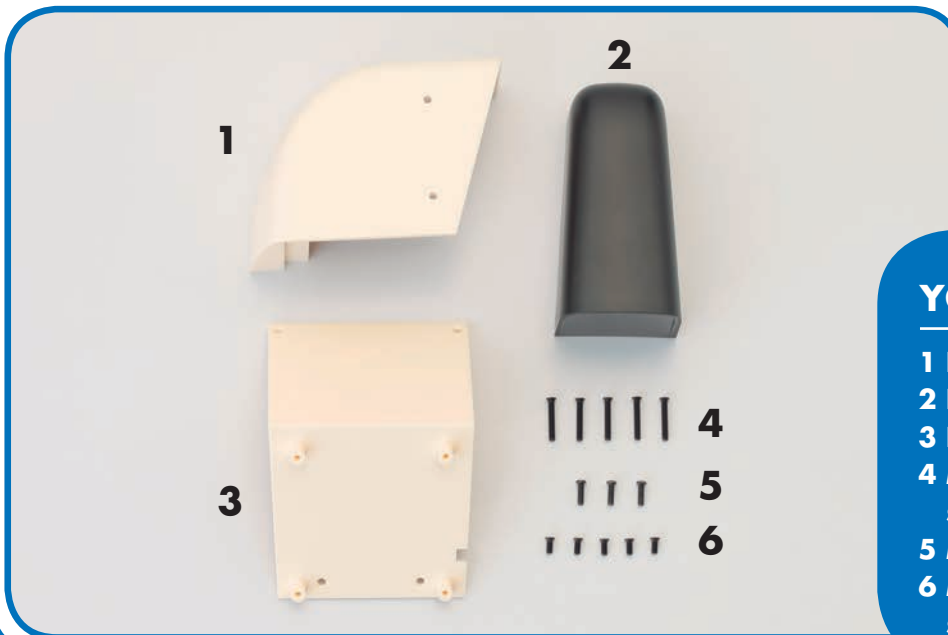
Complete the neck stand, then fasten Robi's scarf and first servo motor ready for the head test in Stage 7.

In this stage, you will assemble the left side of the neck stand, then join this with the rest of the parts assembled in previous stages. You will then fit Robi's scarf, and the servo you modified in

the previous stage, to the top of the stand. As this is a temporary assembly, be careful not to overtighten any of the screws, as they will be used again in later stages.



PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Left chest (for neck stand)
- 2 Left shoulder (for neck stand)
- 3 Back (for neck stand)
- 4 M2 x 12mm countersunk screws (x 5)
- 5 M2 x 6mm pan-head screws (x 3)
- 6 M2 x 4.5mm countersunk screws (x 5)

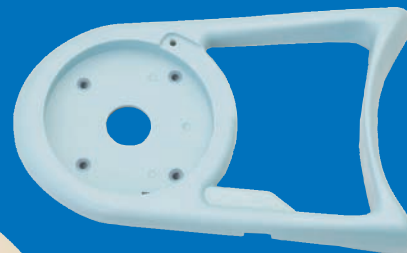
One of each screw is a spare

SAVED PARTS

For this stage, you will need some of the parts you assembled or saved from previous stages. Be sure to have these to hand before proceeding.



Right shoulder and chest assembly (Stage 3)

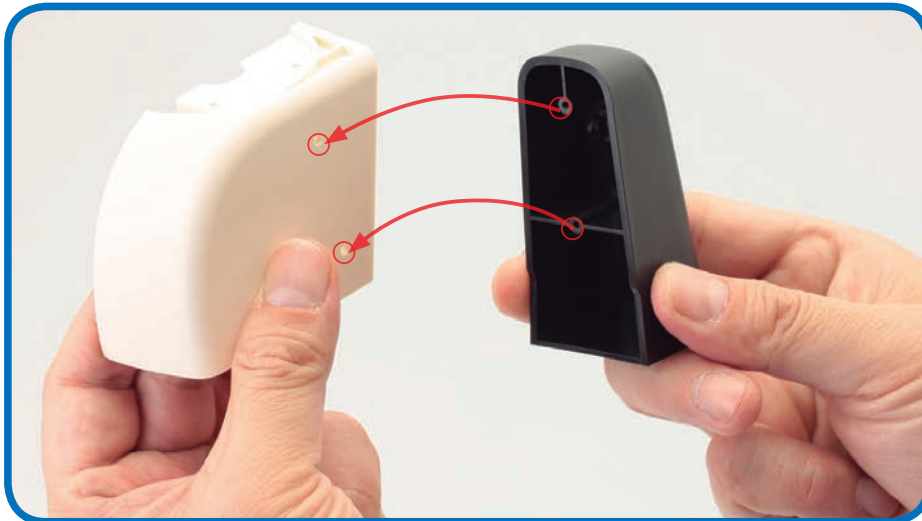


Robi's scarf (Stage 3)



Servo motor (Stage 4)

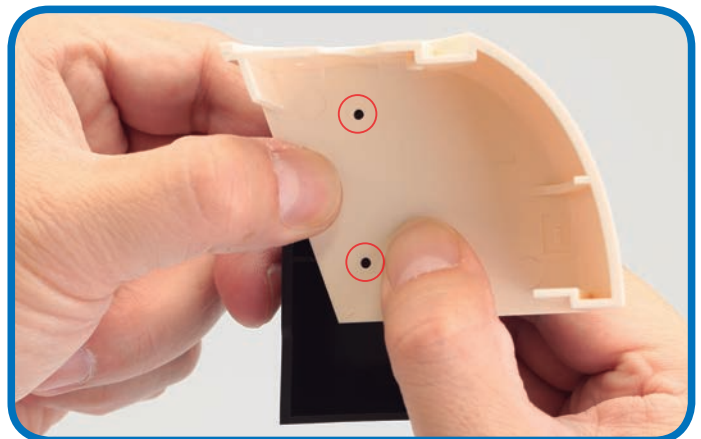
THE CHEST AND LEFT SHOULDER



1 The chest and left shoulder will be assembled in exactly the same way as you did for the right hand side in Stage 3. First, hold the parts as shown and line up the holes marked in red.



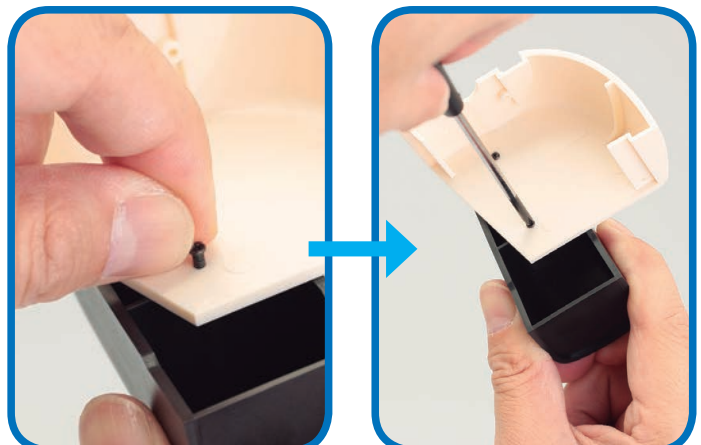
2 Fit the parts together so that the holes are touching.



3 Turn the parts around together so you can access the holes from the inside of the chest.



4 Place an M2 x 6mm pan-head screw into the first hole. Tighten with the screwdriver, but be sure not to overtighten as this can damage the parts.



5 Place a second M2 x 6mm pan-head screw into the next hole, again making sure not to overtighten.

THE NECK STAND



- 6** Take the right shoulder and chest assembly from Stage 3 and hold it against the parts assembled in the previous steps.



- 7** Align the parts so that they fit comfortably together.



- 8** Holding the neck stand together, line up Robi's scarf with the circular recess at the top of the stand.



- 9** Place the scarf into this recessed section. Hold it in place with your finger.



- 10** Still holding the neck stand and scarf in place, take the servo motor that you modified in Stage 4 and lower its servo cable through the central hole of the scarf.

THE SCARF



- 11** Lower the servo into the circular recess in the top of the scarf, then turn the servo mounting base until its four holes line up with those of the scarf (see circles).



- 12** Place an M2 x 12mm countersunk screw into one of the holes and tighten carefully.



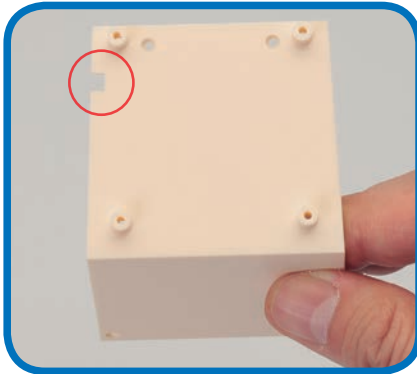
- 13** Repeat with three more M2 x 12mm countersunk screws to attach the mounting base in each of its four holes.



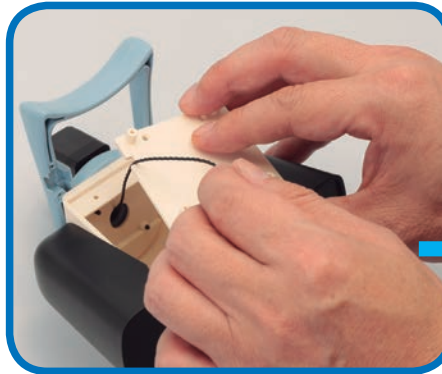
Your assembly should look like this.



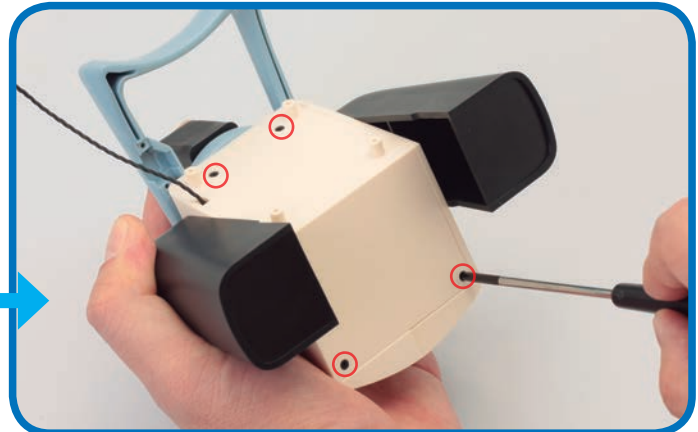
ATTACHING THE BACK



- 14** Inspect the neck stand's back. There is a rectangular cutaway in the top left corner (circled).



- 15** Lay the neck stand assembly face down on your work surface. Then, holding the servo cable so that it protrudes through the rectangular cutaway, enclose the neck stand by fitting the back as shown.



- 16** To complete the neck stand, tighten four M2 x 4.5mm countersunk screws into the holes circled in the photo above right.

Assembled neck stand and servo

This stage is complete, and the mechanics of Robi's head movement are beginning to take shape, ready for the test in Stage 7.



COMING IN THE NEXT STAGE



Batteries not included

The battery pack that will power Robi's servos.

STAGE 6: TESTING ROBI'S HEAD SERVO

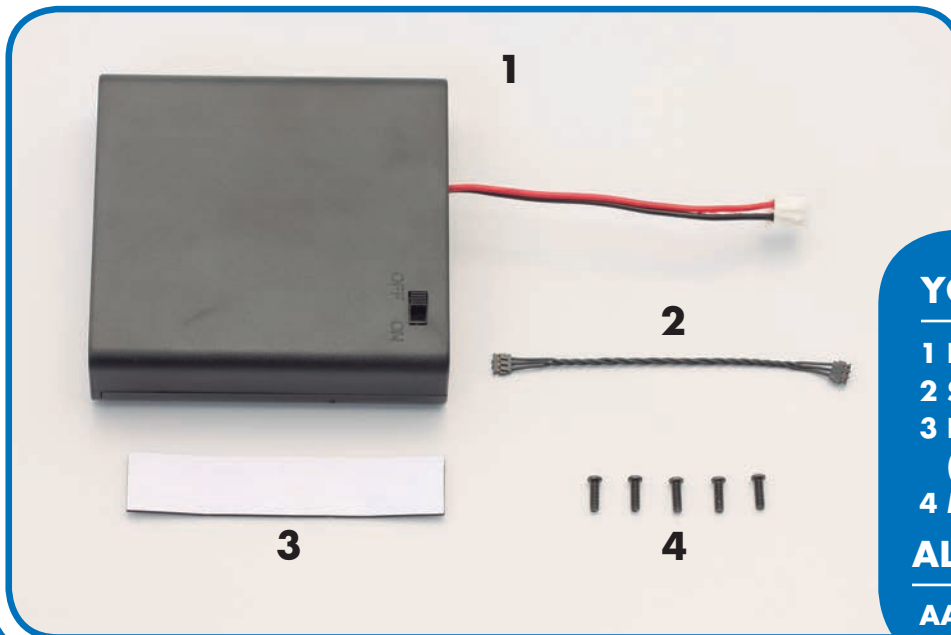
In this assembly, you will attach the test board and battery pack to the neck stand assembled in previous stages, before connecting and testing the first of Robi's servos.

By the end of this stage, you will see the first of Robi's moving parts in action, when you test the rotational movement of the servo that will turn his head. The work is not difficult, but

be very careful when handling the electronic parts, as the exposed components are delicate and easily damaged. You will also need four AA batteries for the battery pack.



**PARTS TO BE
ASSEMBLED**



YOUR PARTS

- 1 Battery pack
- 2 Servo cable (70mm long)
- 3 Double-sided adhesive pads (4 pieces)
- 4 M2 x 6mm pan-headed screws (x 5)

ALSO REQUIRED

AA batteries (x 4)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



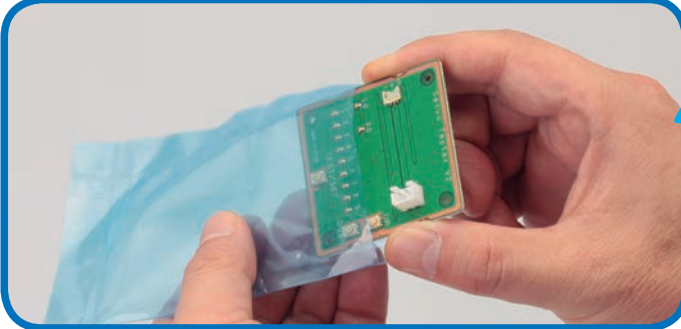
Neck stand assembly (Stage 5)



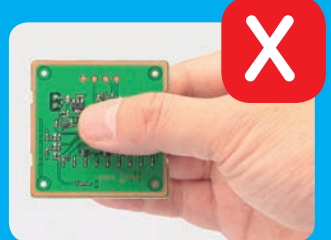
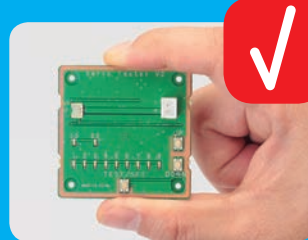
Test board (Stage 3)

THE TEST BOARD

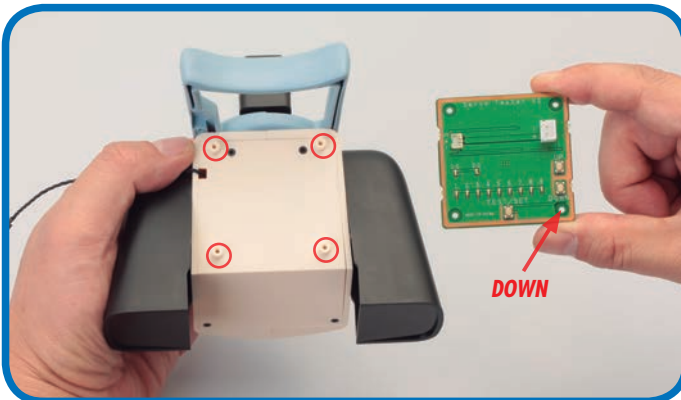
TIP!



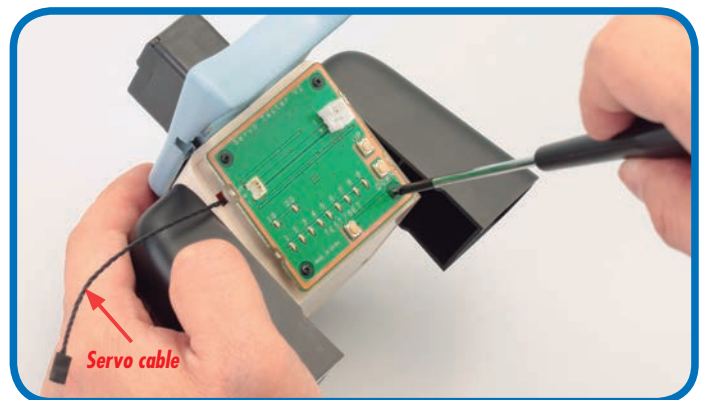
- 1** Carefully remove the test board from its plastic bag. Do not touch any of the electronic components.



Always handle the test board by holding the sides, making sure not to touch any of the components on its surface.

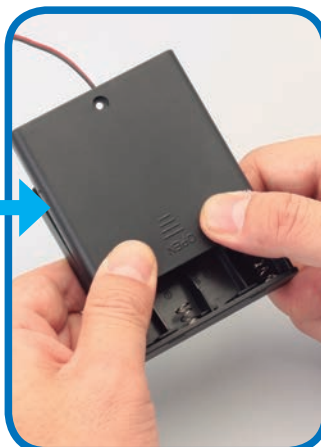


- 2** Holding the parts as shown (with the word DOWN at bottom right, as arrowed), fit the board onto the back of the neck stand, lining up the four circled corner holes.

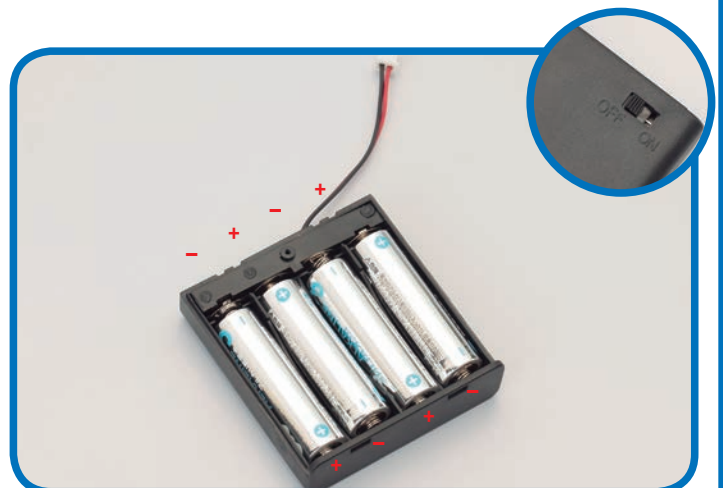


- 3** Ensure that the servo cable (arrowed) is not trapped under the board. Insert M2 x 6mm pan-headed screws into each of the four corner holes and gently tighten them.

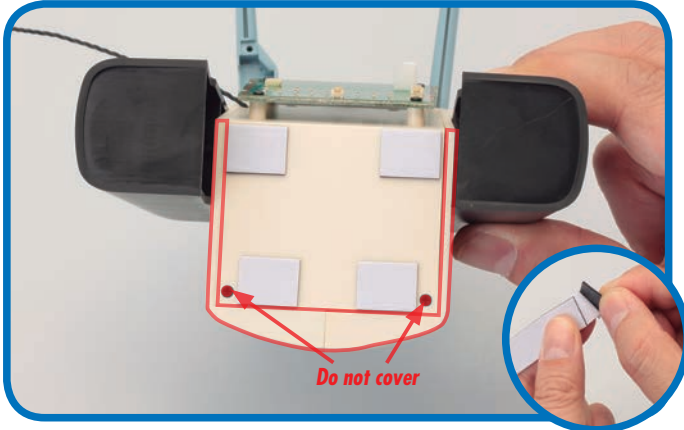
THE BATTERY PACK



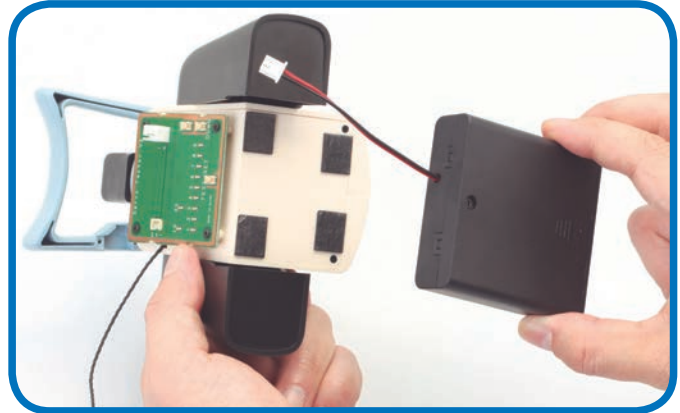
- 4** Take out the screw holding the cover of the battery pack in place and slide the cover off.



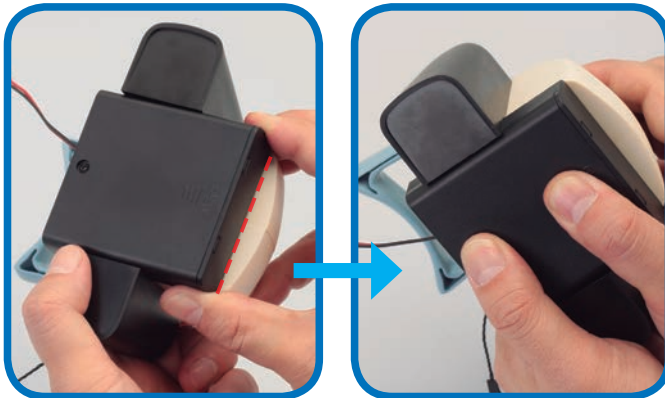
- 5** Make sure the power switch is set to OFF. Fit four AA batteries into the pack with their positive and negative ends positioned as shown, then replace the cover.



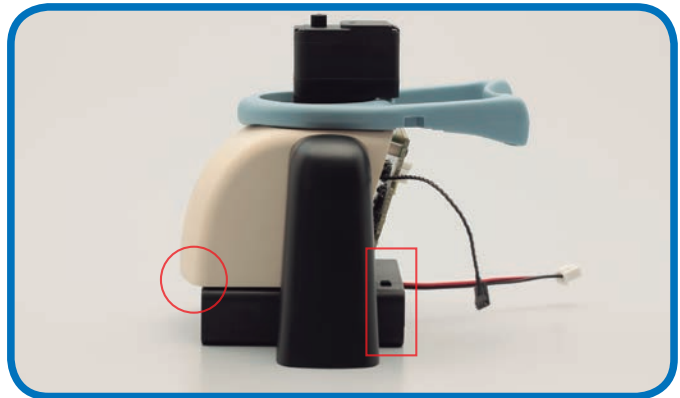
- 6** Peel the four double-sided pads off their backing. Stick them under the neck stand as shown, being sure not to cover the two holes at the front.



- 7** Peel the white protective backing off the four adhesive pads, then align the battery pack with the cable pointing towards the circuit board and the screw facing you.

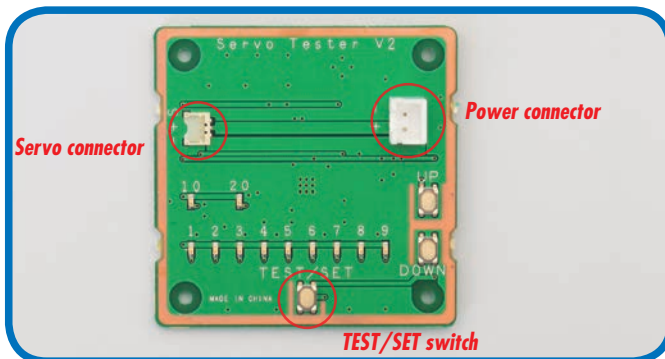


- 8** With the battery pack in line with the ridge running along the front of the neck stand (see dotted red line) press it firmly into place.

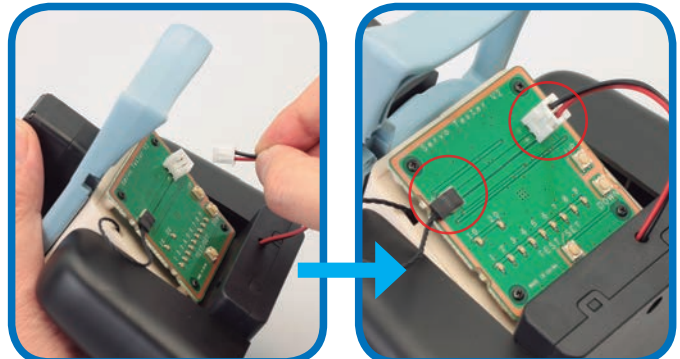


- 9** Your assembly should look like this, with the front of the chest hanging over the edge of the battery pack and the battery pack protruding beyond the rear of the stand.

THE TEST BOARD



- 10** Familiarise yourself with the connectors and switches on the test board. There are two connection points: the one to the left is the servo connector, and the other, larger one is the power connector. At the bottom is the TEST/SET switch.

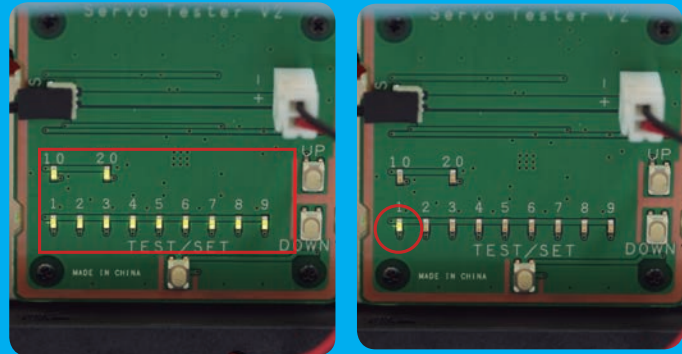


- 11** Plug the loose servo cable into the servo connector and press gently on the protective pad on top in the same way you did when connecting to the servo itself. Then push the cable from the battery pack into the power connector.

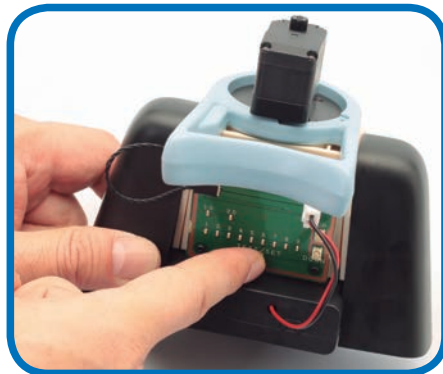


- 12** Push the switch on the battery box to ON. All the LEDs on the test board should flash twice, then the one under the number 1 should remain lit.

HELP!



If the LEDs do not light up, check that the power cable is connected properly. Additionally, if any one of the batteries is aligned the wrong way within the pack, it will cause the test to fail.

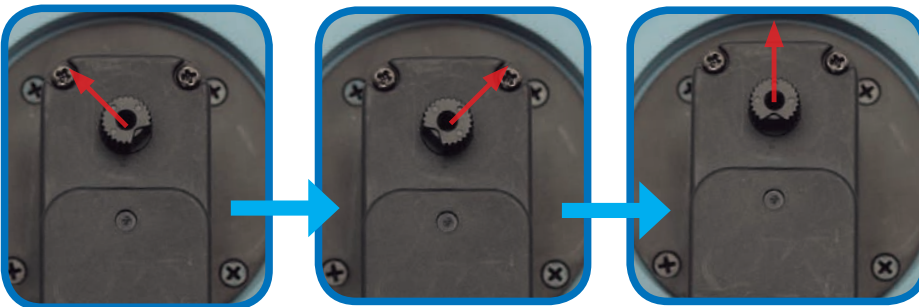


- 13** Press the TEST/SET switch briefly to run the servo test. The shaft at the top of the servo casing should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. After you have run the test, turn off the power switch on the battery pack.

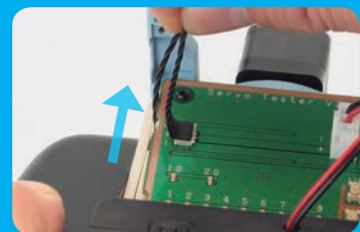
HELP!



If the LEDs flash rapidly, you have a faulty connection. Double-check the servo cable's connection at the test board, turning the power off before you do. If the light still flashes, carefully unscrew the servo housing and check that the internal connector plug is in place. If you pressed the wrong button, reset the board by holding down the TEST/SET switch for several seconds.



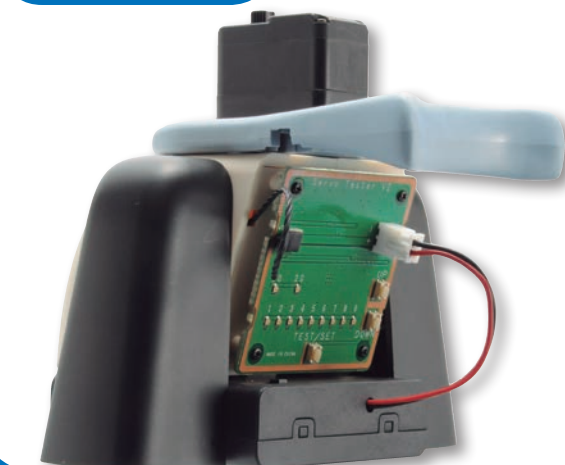
BE CAREFUL!



If you have to remove the servo to check the connection, disconnect it from the test board with extreme care by pulling the servo cable very gently and directly upwards away from its connector.

Assembled neck stand

This stage is complete. Make sure the power is turned off, and store the neck stand carefully without touching any of the electrical components. In the next stage, you will attach Robi's head to the neck stand and test its movement.



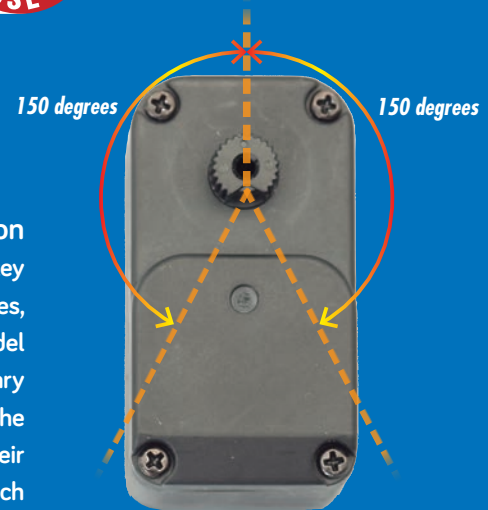
ROBI'S SERVO MOTORS



Robi is fitted with 20 servo motors to control all his movements. This is a brief introduction to their roles.

Servo rotation

Servo motors (or simply 'servos') are key components in a wide range of electronic devices, and are commonly used in remote-control model cars and planes, as well as robots. Unlike ordinary electric motors, which spin continuously when the power is applied, servos are designed so that their shaft only turns through a set, small angle – such as that found in a human joint.



The servos fitted to Robi are Futaba RS308MD models. These have a maximum rotational range of 300 degrees, and are able to turn 150 degrees in either direction from the central rest point, as shown above.



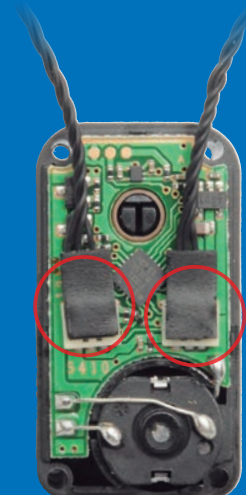
On completion, Robi's head will be able to turn in three dimensions, allowing him to nod, shake his head and tilt it from side to side.

Degrees of freedom

Robots' agility is often measured in 'degrees of freedom', and this term typically relates to the number of servos they have. Each servo rotates back and forth around a single axis, giving each a single independent movement – for example, allowing a joint to be angled backwards or forwards. This is counted as one degree of freedom. Robi's head and neck will eventually be fitted with three servos, giving his head three degrees of freedom, shown by the arrows on the photos to the left.

Wiring a servo

Each servo needs to be connected to the microcomputer board inside Robi's body, which supplies the electricity needed to power its rotation and tells it what action to perform – when to turn, and how far. However, not all of Robi's 20 servos need to be individually connected to the board, as this would mean a mess of wires and require 20 connections on the board itself. Instead, it is possible to wire related servos (such as those in each leg) together in a sequence. This is sometimes called a 'daisy chain'. The power and control information is carried from the board to the first servo, along to the next, and onwards until each individual unit is connected in the chain.



The servos used in the Robi assembly each have two connection points to allow them to be wired in a 'daisy chain'.

COMING IN PACK 3

Your next **FOUR** complete Stages, as Robi starts to come to life!

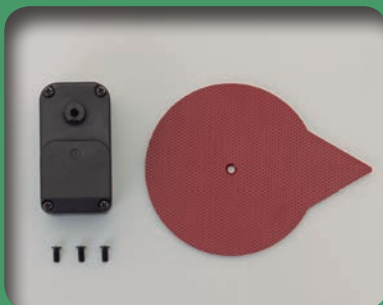


Stage 7

Test Robi's head movement using the neck stand

PARTS PROVIDED

More parts of Robi's head, and the left forearm panel.



Stage 8

Add the ear trim panels to Robi's head

PARTS PROVIDED

Another ear panel, allowing you to close up Robi's head, and the second of Robi's 20 servos.



Stage 9

Begin assembling Robi's right arm, starting with the forearm

PARTS PROVIDED

The right forearm frame used to join Robi's arm servos, and the first parts of his hands.



Stage 10

See the first of Robi's moving parts in action, as you test his head servo

PARTS PROVIDED

The right upper arm frame used to join Robi's arm and shoulder servos, and another servo cable.